Margaret Ellen Seehorn, University of Michigan CSE PhD

Academic Statement of Purpose

I aim to study, develop, and deploy **intelligent human-AI interactive systems that dismantle systemic inaccessibility** in both the real and digital world**, using co-design** to ensure these systems not only serve but honor their historically under-served users.

I am motivated to pursue this line of research because, despite ⅙ of the global population experiencing disability [2], the physical and digital world are built without prioritizing disabled needs, stifling countless people's ability to fully engage with their lives and communities. Technology can be either a great equalizer—mitigating the obstacles of systemic ableism by providing new modalities and mechanisms for interaction—or a sophisticated instrument of exclusion—encoding bias and creating new structures to manifest inaccessibility. While the route of sophisticated exclusion is not taken solely out of malice, but also out of naive underconsideration, the effects are the same nonetheless. In an increasingly digital climate where algorithmic decision making and AI-integration are evermore ubiquitous and public adoption of novel tools is unprecedentedly fast, the stakes of our technological choices have never been higher. The "great equalizer" path requires intention and commitment.

A CSE PhD at Michigan will be the start of the greatest commitment I can make: a life's work. Here, I will be empowered to catalyze systemic change in accessibility through my graduate research and preparation for the career that follows. I will develop the expertise needed to become an accessibility-oriented professor and researcher at an R1 institution. There, via research, mentorship and teaching, I will support future technologists in prioritizing accessibility in their work, be it in academia or industry.

I have spent my undergraduate years preparing for this career as a researcher and educator. I selected my majors with accessibility research in mind—Sociology, through its emphasis on institutional structures and potential for person-level analysis equips me to understand and elevate the lived disabled experience, while the technical skills afforded by Computer Science prepare me to reflect those perspectives in deployed systems. Further, I have been a teaching assistant in four distinct Computer Science courses from the introductory sequence to an upper-division systems class, which has provided me the opportunity to develop skills in inclusive teaching practices that foster student self-efficacy. Finally, I have undertaken three research experiences: each instrumental in defining my research interests and philosophy.

In my first research experience, under **Jennifer Mankoff** at University of Washington, I explored using machine embroidery for tactile graphic (TG) production. Traditional embossed TGs lack sufficient textural diversity to convey complex information, so I implemented metaheuristic optimization to assign maximally-distinct embroidery textures—which have greater tactile diversity—to create a TG based on an SVG. Beyond technical skills, this work taught me to prioritize early human-centered evaluation. Our approach involved a literature search, then system building, and user studies only after implementation. Our studies affirmed embroidery could yield superior tactile distinction, leading to a 2022 ACM SCF demo [1], but we also received feedback about desires for embroidered TGs that, had we known earlier, would have addressed real user needs rather than anticipated ones.

With this user-centered lesson in mind, I entered my second REU. At Carnegie Mellon HCII, under **Vincent Aleven** and **Haiyi Zhu**, I focused on teacher-driven integration of IEPs (Individualized Education Programs) into ALTs (Adaptive Learning Technologies) in Special Education. ALTs use AI to personalize learning, but are rarely designed for disability. IEPs contain vital context: accommodations and individualized learning goals. IEP-ALT integration could improve classroom accessibility. To explore integration possibilities, I designed and conducted semi-structured interviews with Special Education teachers. This work, for which I was named a Goldwater Scholar, culminated in design recommendations for IEP-aware ALTs: bridging a literature gap between cognitive accessibility and educational technology.

This project enlightened me to AI's growing ubiquity and, thus, its expanding capacity to exacerbate harm or be leveraged for equity. I realized that, to pursue technology as "the great equalizer", my approach to accessibility must address AI: pitfalls and promises alike. This project shaped not only my interests, but also my research philosophy. I take pride in my work's potential to impact millions of students, but impact is conditional on implementation. If our recommendations never influence the tools themselves, those whose experiences inform my research never benefit: an inequitable end. Qualitative methods are indispensable, yet, it is my responsibility as a researcher to leverage technical skills to reify qualitative knowledge in tangible outcomes. For my work, mixed-methods is an ethical imperative.

This philosophy permeated my third research experience at University of Michigan, where, under **Anhong Guo**, I was able to more deeply understand my newly identified research focus: accessibility through an AI lens. I studied AI-based mobile assistive technologies for blind users: leveraging mobile automation to enable add-ons like camera-aiming assistance and cross-checking AI outputs to help bridge the gap between academic research and commercially available tools. I utilized iterative co-design with two blind participants, with whom we ascertained user needs, expectations, and envisioned usage to inform the selection, development, and evaluation of add-ons. This work was not without challenges—our in-the-wild deployment and testing plans were foiled by interfacing with third party softwares with their own accessibility issues. Still, co-design satisfied my need to center the people my research aims to help at every stage: preserving their agency by designing with, not just for, them and ensuring their contributions do not go undervalued via rapid technological returns.

These experiences shaped my research interests and philosophy, pointing me towards the University of Michigan Computer Science & Engineering PhD. Here, I could **deepen my expertise in both Accessibility and Computing and Human-Computer Interaction**, while broadening my knowledge in other subject areas to supplement my ability to do exciting, high-impact work at the intersection of Accessibility and HCI. Michigan's faculty includes multiple professors I see as outstanding potential mentors. I am drawn to **Anhong Guo** for a number of reasons. His expertise in human-centered accessible system design and AI systems would naturally support me in continuing my exploration of in-situ mobile AI for accessibility, considering that investigation began in his lab. Continuing to work with him at the graduate level would afford me the opportunity to gain depth in areas I found particularly fascinating, like multi-model verification and debugging of answers to visual questions generated by large language models, as well as breadth in areas he focuses on that I was not able to robustly explore in my initial project, such as Augmented Reality. This breadth would allow me to explore AI use for accessibility using a wider variety of methods of information presentation—such as embedding information directly into the interface in use as opposed to taking the interface as semi-static input—, through a wider variety of devices for engagement—such as through expanding from smartphone focus to one that includes wearables—, or both. Furthermore, I envision productive graduate study under **Dhruv Jain**. I met members of his team at ASSETS, and was deeply impressed by the work they do at the intersection of AI and accessibility for the Deaf/Hard Of Hearing community—work that is particularly special to me given my own membership in that community. I envision my existing experience with human-centered methods as well as language models would help me effectively contribute to projects that extend works like CartGPT. Further, with Professor Jain's guidance, I could extend the scope of this valuable accessibility augmentation to greater embodiment through work in mixed and augmented reality settings. I am also excited by the prospect of collaborating with **Vera Liao** and **Nikola Banovic**, whose research on AI trust and explainability complements my focus. Professor Liao's work in uncertainty communication would inform verification and accuracy feedback in accessible AI systems, while Professor Banovic's focus on explainability for users with limited technical expertise would enhance AI-driven accessibility tools. While neither has a

specific focus on accessibility, their expertise is critical to my envisioned applications, and their guidance would be invaluable to my growth as a researcher.

Working with any of these professors will prepare me to use human-centered methods to study, develop, and deploy systems for usable accessibility. However, Michigan's program excites me beyond these professors. In particular, the AHA group at University of Michigan offers a unique chance to be surrounded by a community of research where the ideas and priorities around accessibility I care so deeply about are not isolated to a specific advisor, but rather, permeate the broader community. This, in combination with the availability of multi-PI research collectives in both Human-Centered Computing and Artificial Intelligence, gives me confidence that my interests in AI-supported, human-centric, accessibility will be well supported in and outside of specific advising relationships. Members of all three of these groups regularly publish at high-impact venues such as CHI, ASSETS, UIST, and AAAI; it is clear that this community of collaboration will help me attain not only the skills, but also the credentials necessary to find success in my eventual career. I know doing a Computer Science PhD at Michigan will empower me to make tangible impacts in accessibility and I am confident that my previous experience has prepared me to embark on this journey. I look forward to the growth I will experience and the contributions I will make if given the opportunity to join the Michigan community.

References: [1] Seehorn & Kim et al. 2022. https://doi.org/10.1145/3559400.3565586 (SCF), [2] 2023. https://www.who.int/news-room/fact-sheets/detail/disability-and-health